SkillBridge AI

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AI-Powered CV Skill Gap Analysis and Personalized Course Recommendations

Homework 1 | Project Selection and Exploratory Data Analysis

GitHub Repository: https://github.com/sudemkalip/SkillBridge_AI

1. Project Name and Overview

SkillBridge AI is a proposed career-support tool for university students and recent graduates. The idea is simple: a user chooses a target occupation and uploads a CV. The future system would identify skills mentioned in the CV, compare them with O*NET occupational requirements, and suggest relevant courses. The proposed labels are Learn (not demonstrated in the CV), Improve (some evidence, but more development may be useful), and Sufficient (evidence seems to meet the target requirement). Beginner courses would be suggested for Learn and intermediate or advanced courses for Improve. This application has not been built yet; Homework 1 focuses on understanding the data.

The project mainly relates to SDG 4, Target 4.4 (skills for employment), and also supports SDG 8, Target 8.5 (career readiness). It does not claim that taking recommended courses guarantees employment.

2. Problem and Research Questions

Students may know which job they want but not which skills matter most for that occupation. A CV also does not directly show how well someone meets a job profile. Meanwhile, online course platforms offer many choices, making it difficult to select a suitable course level. We want to connect these three pieces of information in a practical way.

- Can relevant skills be identified from a CV using NLP?
- How can CV evidence be compared with O*NET requirements for a selected occupation?
- How should skill gaps be explained without treating a missing CV mention as lack of ability?
- Can course skills and difficulty levels support useful learning recommendations?

3. Data Sources And Scope

O*NET Database 31.0 (official): Occupation Data, Essential Skills, and Knowledge. Occupation codes can connect job titles to skill and knowledge requirements. Importance scores describe occupations, not a particular student.

Coursera Course Dataset (azrai99, Hugging Face; third-party copy): The 2024 course file includes titles, descriptions, skill tags and difficulty levels. It is not an official Coursera data release. The analysis used the Coursera data in the working Colab environment; the course file was reported as CSV in the final notebook workflow. The report uses the recorded EDA results, not a new download of the dataset.

No personal CV data was used for Homework 1. CV extraction, skill matching and course recommendations are planned for later stages.

4. Tools and Initial Preprocessing

Exploratory analysis was carried out in Google Colab using Python, pandas and Matplotlib. The source tables were inspected for dimensions, missing values and exact duplicate rows. We created working copies, removed duplicate rows where applicable (none were found), removed the extra "Unnamed: 0" column from the Coursera table, and trimmed spaces in selected Coursera text fields. Missing course levels were kept rather than guessed. We did not complete skill-name standardization, occupation-to-course joins, or model training in this assignment.

Reproducibility Note: The GitHub notebook and the report should be checked together before submission. The source files must be uploaded to the notebook's expected paths, and the CSV/Excel extensions in the notebook must match the files actually used. The statistics and figures below come from the completed Colab analysis and its saved outputs/screenshots.

5. Exploratory Data Analysis

5.1 Dataset Overview

Dataset	Rows	Columns	Missing cells	Duplicate rows
O*NET Occupation Data	1,016	3	0	0
O*NET Essential Skills	18,200	15	9,100	0
O*NET Knowledge	60,060	15	99,416	0
Coursera Courses	6,645	14	8,526	0

Table 1. Counts refer to the original tables before removal of the extra Coursera index column. "Missing cells" is not the same as the number of unusable records. After basic cleaning, the Coursera table had 13 columns; row counts stayed unchanged.

5.2 O*NET: Average Skill Importance

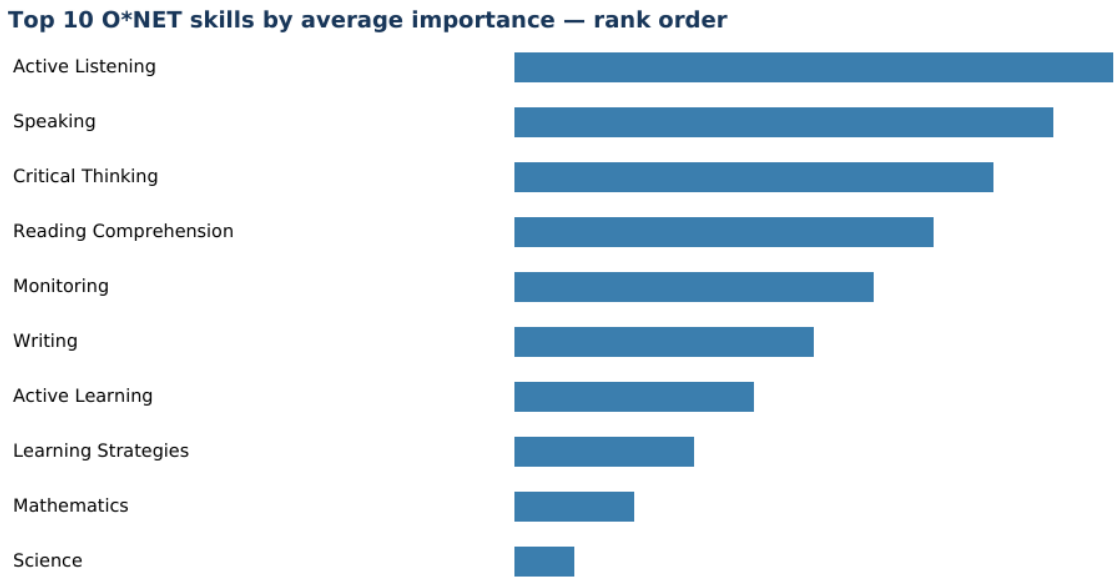


*Figure 1. Rank-order summary of average O*NET skill importance (reconstructed from EDA results; bars indicate rank, not exact scores).*

Active Listening, Speaking, Critical Thinking and Reading Comprehension are among the highest-ranked skills in the cross-occupation average. This helps us understand the scale, but a future recommendation must use the requirements of the user's chosen occupation instead of these overall averages.

5.3 Coursera: Frequently Listed Skills

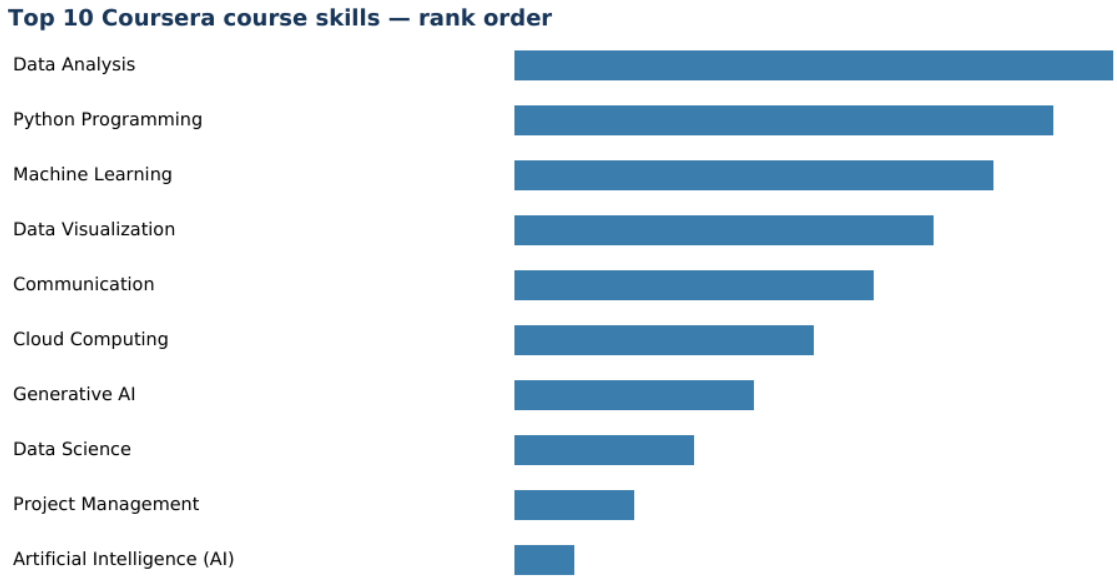


Figure 2. Rank-order summary of frequently tagged Coursera skills (reconstructed from EDA results; bars indicate rank, not exact counts).

Data Analysis, Python Programming and Machine Learning appear frequently in the course skill tags. A single course can list several skills, so the bars count skill mentions, not unique courses. The course vocabulary is often more specific than the broad skill labels in O*NET; later matching will need more than exact word comparison.

5.4 Coursera: Course Level Distribution

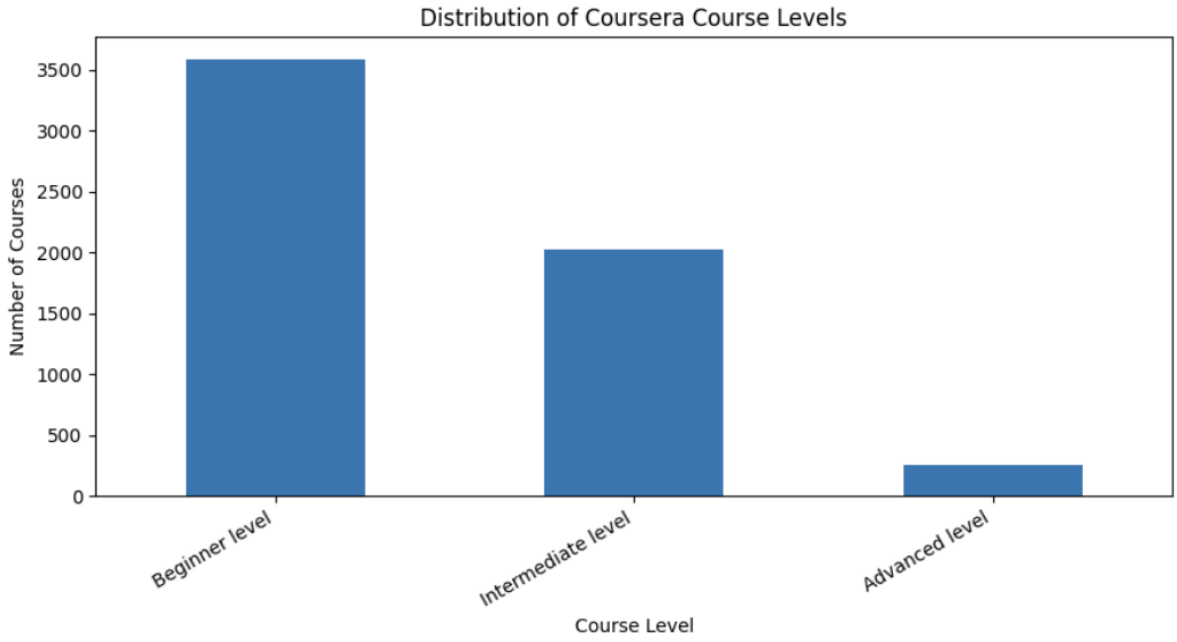


Figure 3. Course levels among records with a reported level (original Colab chart).

Beginner courses are the largest group, followed by Intermediate courses. Advanced courses are much less common. The Level field is missing for 778 of 6,645 course records (11.7%). We should not automatically assign those courses a level. The uneven distribution may also limit choices for users who need advanced courses.

5.5 Coursera: Missing Values

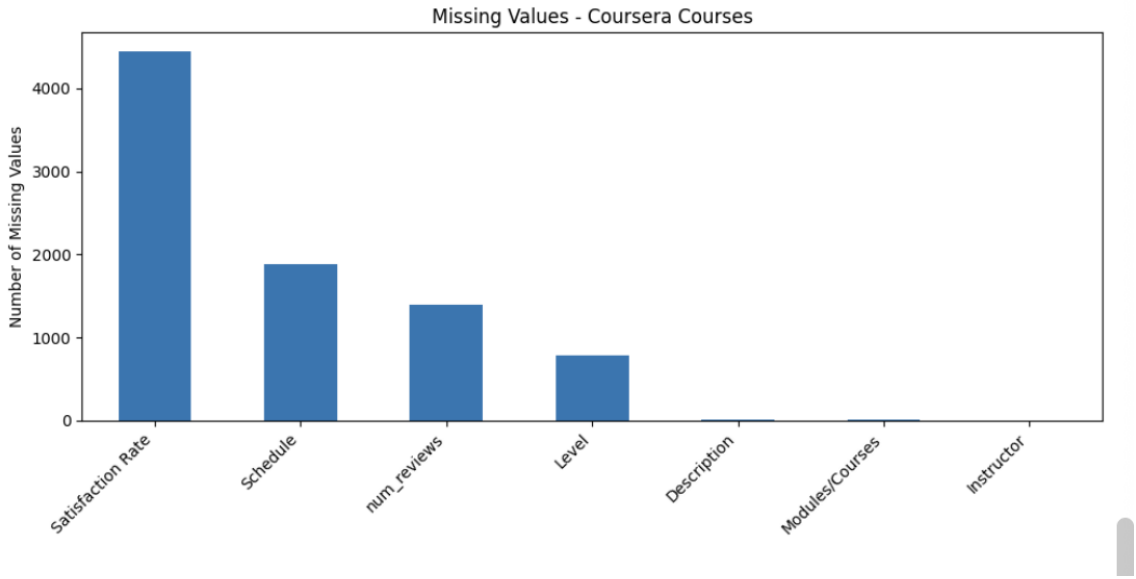


Figure 4. Missing values by field in the Coursera course data (Colab output).

Satisfaction Rate, Schedule and num_reviews have many missing entries. The recommendation-relevant Level field has 778 missing values, while Description has 10. These gaps matter because difficulty and course descriptions are useful when ranking courses. Missing fields were identified but not filled with invented values.

5.6 O*NET Knowledge: Missing Values

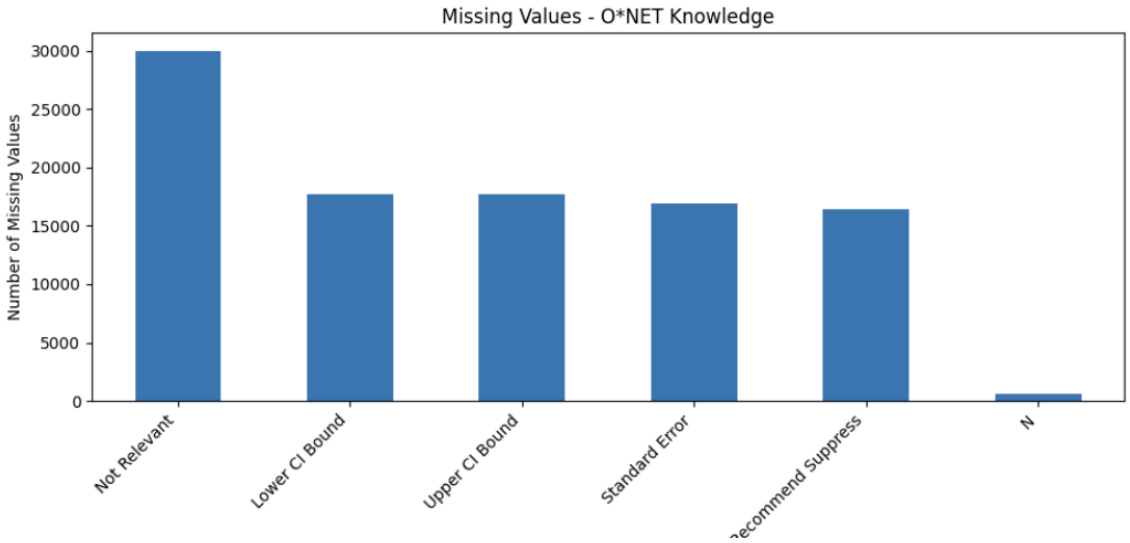


*Figure 5. Missing values by field in the O*NET Knowledge table (Colab output).*

Missing entries are concentrated in fields such as Not Relevant, confidence interval bounds, Standard Error and Recommend Suppress. Many are statistical or relevance-related fields rather than missing occupation names. For that reason, deleting every row containing a blank cell would be a poor first cleaning step. The intended use of each field should be checked before filtering.

6. Main Findings and Implications

The initial EDA suggests that both datasets can support the SkillBridge AI idea. O*NET offers structured occupational requirements, while the course dataset offers learning resources with skill tags and levels. The main difficulties are incomplete course metadata, uneven course-level coverage, and differences in the way skills are named. A further limitation is that a CV provides evidence of experience but cannot measure a person's ability perfectly.

In later stages, we plan to join O*NET information by occupation code, map related skill terms, extract candidate skills from CV text, and let users correct the extracted information. Course recommendations would then use both skill relevance and course level. These steps are proposed, not completed results.

7. Planned Method, Evaluation and Ethics

The proposed application would extract text from PDF or DOCX CVs, use NLP to identify skill evidence, compare it with the selected occupation, and recommend courses. Possible evaluation measures include precision/recall for extracted skills and manual review of the relevance of suggested courses. Users should be able to edit their profiles. Sensitive CV details should not influence recommendations, and unnecessary personal data should not be stored.

8. Conclusion

For Homework 1, we selected a project linked to SDG 4 and SDG 8, examined four public-data tables, completed basic cleaning and missing-value checks, and interpreted five EDA visualizations. The results helped us see both the potential of the data and the issues that must be addressed before building a reliable recommendation system. The AI application itself remains future work.

9. References

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- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. EMNLP-IJCNLP.